SORh

Hyperbolic relaxation methods for elliptic problems in computational fluid dynamics

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M.S. Scientific Computing & Applied Mathematics • UC Santa Cruz

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Roadmap

→ How can a hyperbolic pseudo-time model improve the performance of a local elliptic solver?

1D FOUNDATION

1. **MODEL** — mixed-derivative formulation
2. **STRUCTURE** — AOR / SOR identity
3. **MECHANISM** — weak-diffusion propagation

2D EXTENSION

1. **MODEL** — directional construction
2. **METHOD** — full update + ordering
3. **RESULTS** — Poisson + nonlinear diffusion
4. **APPLICATION** — FLASH self-gravity

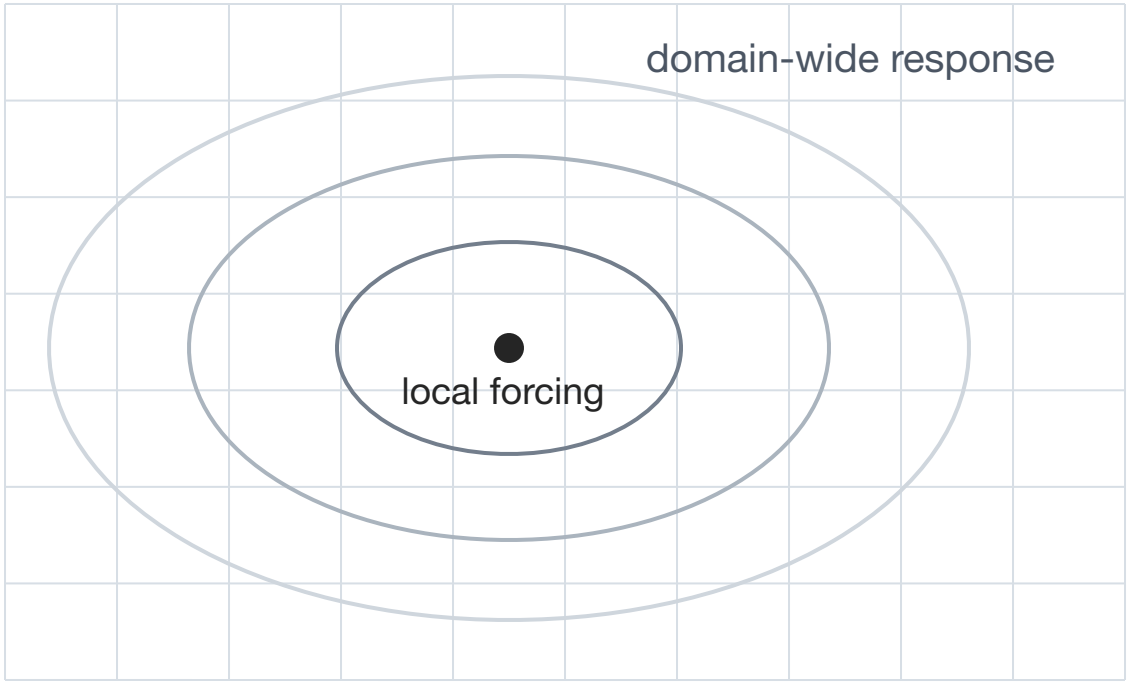
Companion application: projection-based divergence cleaning in a high-order MHD scheme

Properties of Elliptic Equations

$$\Delta\varphi = f$$

sparse local stencil

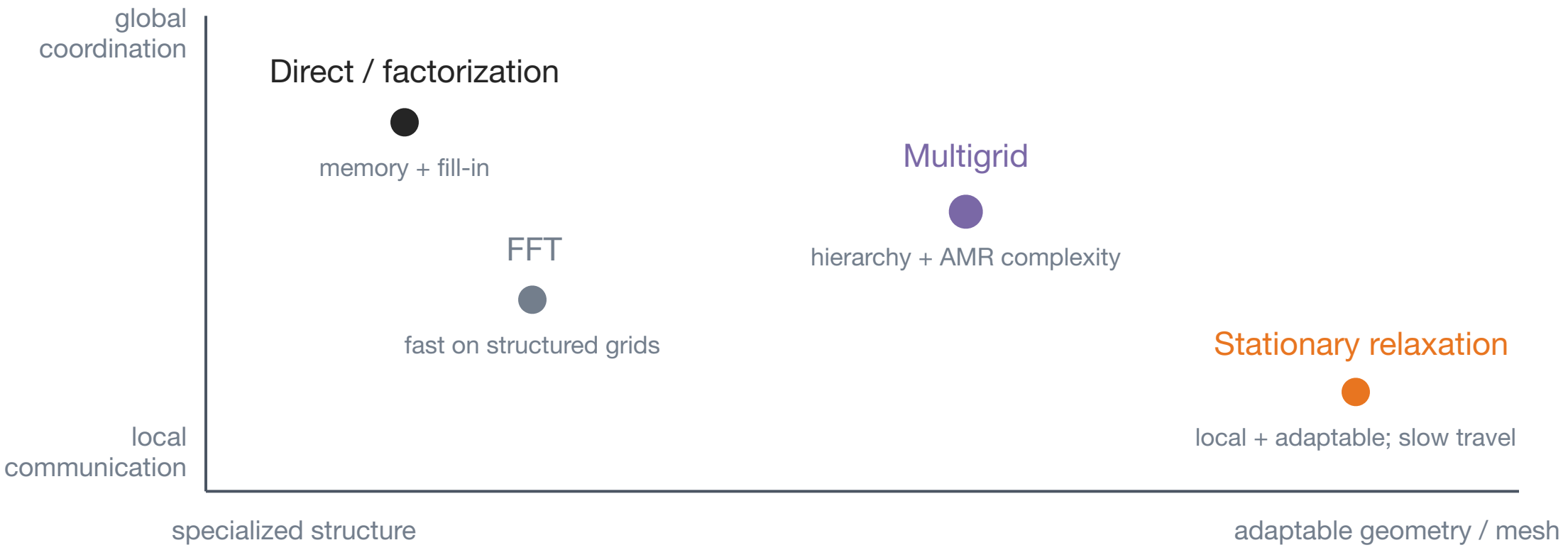
- Well-posedness - unique under standard assumptions
- Regularity - smoothness supports finite-difference analysis
- Globally dependent - local changes influence the entire solution



→ Finite-difference elliptic matrices are locally sparse but globally coupled

→ Direct factorization is typically expensive

Qualitative Tradeoffs for Elliptic Solvers



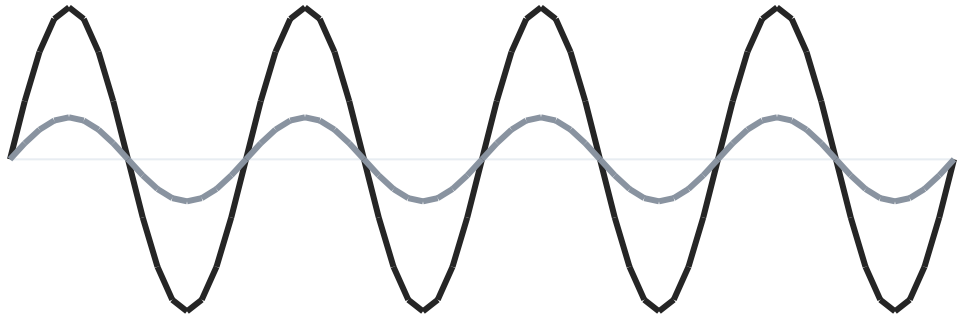
Production reference: multigrid

Primary controlled comparison: classical relaxation

Diffusion damps oscillatory error quickly, but smooth error slowly

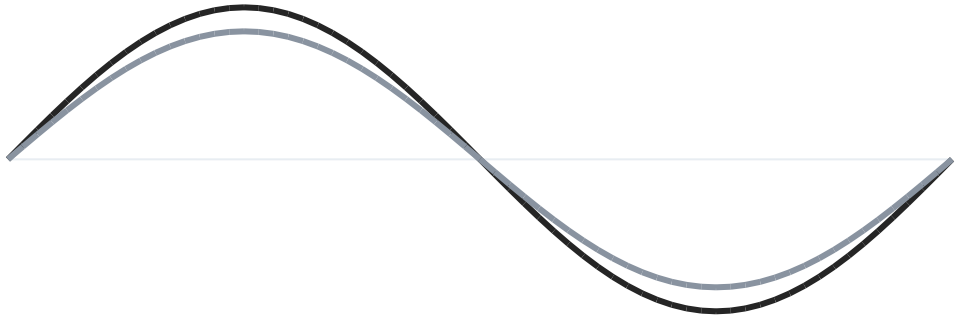
$$u_t = -Au + f \quad \rightarrow \quad e_t = -Ae$$

Oscillatory mode



rapid damping

Smooth mode



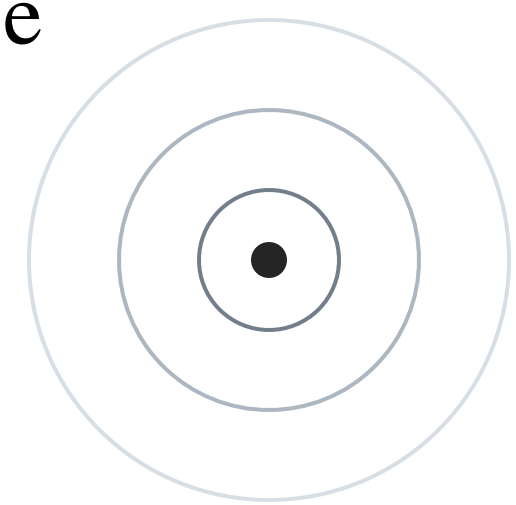
slow decay

before after

Diffusion vs. Directional transport for moving error information

DIFFUSION

$$e_t = \Delta e$$



diffusive propagation

DIRECTIONAL TRANSPORT

$$e_t = e_x$$



movement along a characteristic

How can an ordered elliptic relaxation method be built around this transport structure?

SORh links a scalar hyperbolic model to the SOR/AOR family

METHOD	CENTRAL CONSTRUCTION	DISTINCTIVE ROLE
STATIONARY ITERATIONS		
Classical SOR	Gauss–Seidel update, weight α	One parameter sizes an ordered correction sweep.
AOR (Hadjidimos)	$r_A\delta_{upstream} + \omega_A R$	Two parameters separate propagation and residual injection; SOR when $r_A = \omega_A$
HYPERBOLIC RELAXATION		
Nishikawa	Evolve u and $q \approx \nabla u$	Hyperbolic dynamics reach a diffusive steady state with $\Delta t = O(h)$
SORh	$u_{tx} = (\nu(u)u_x)_x$	Scalar ordered sweep; its linear form maps to AOR and to SOR at unit CFL.

One-dimensional problems isolate what nonlinearity changes

NONLINEAR PROBLEM

$$\frac{d}{dx}\nu(u)\frac{du}{dx} = 0, \quad \nu(u) = \frac{u}{1+u}$$

implicit steady solution

$$u - \ln(1+u) = (1-\ln 2)x$$

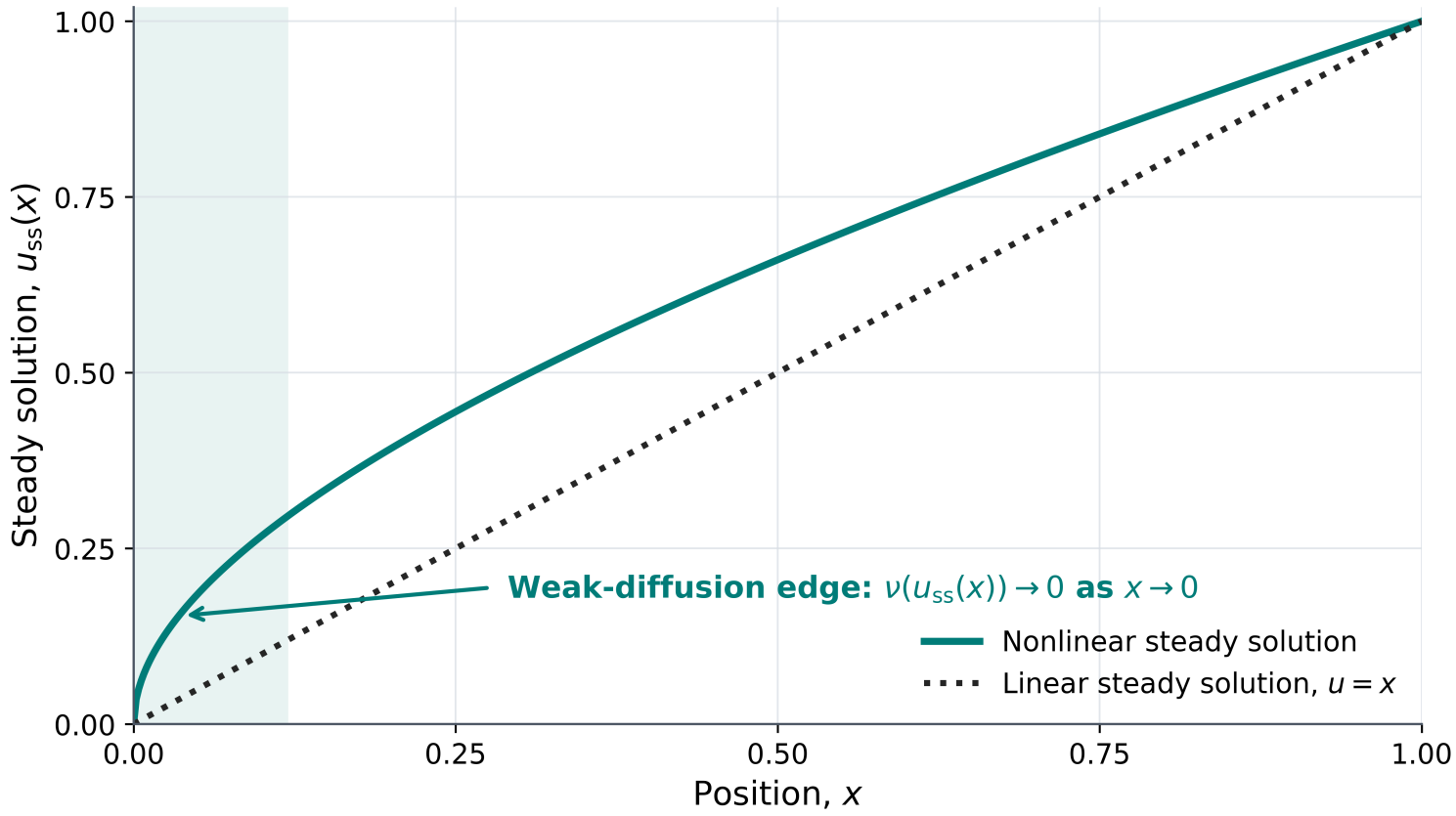
LINEAR REFERENCE

$$\nu_0 u_{xx} = 0$$

steady solution

$$u(x) = x$$

Weak local diffusion = low conductivity near one edge, not weak diffusion everywhere.



Parabolic and hyperbolic models share linear and nonlinear steady states

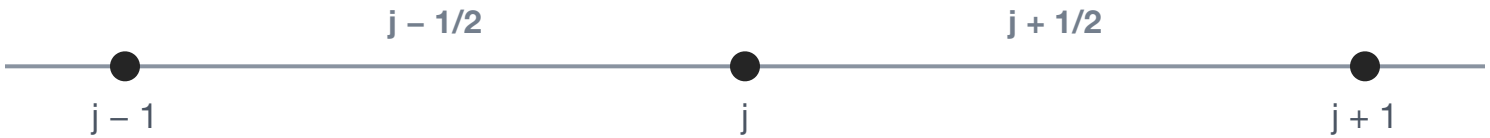
PROBLEM CLASS	PARABOLIC	HYPERBOLIC	SHARED STEADY EQUATION
NONLINEAR $\nu = \nu(u)$	$u_t = (\nu(u)u_x)_x$	$u_{tx} = (\nu(u)u_x)_x$	$(\nu(u)u_x)_x = 0$
LINEAR REFERENCE $\nu = \nu_0$	$u_t = \nu_0 u_{xx}$	$u_{tx} = \nu_0 u_{xx}$	$\nu_0 u_{xx} = 0$

The pseudo-time derivative changes the transient dynamics, but the target steady problem is shared.

SORh: Nonlinear & Linear Discretization

Averaged interface values:

$$\nu_{j+\frac{1}{2}} = \nu\,\frac{u_{j+1}^{n} + u_{j}^{n}}{2}, \qquad \nu_{j-\frac{1}{2}} = \nu\,\frac{u_{j}^{n} + u_{j-1}^{n}}{2}.$$



$$\frac{u_{j}^{n+1} - u_{j-1}^{n+1} - u_{j}^{n} + u_{j-1}^{n}}{\Delta t_H \Delta x} = \frac{\nu_{j+\frac{1}{2}}^{n}\,u_{j+1}^{n} - u_{j}^{n} - \nu_{j-\frac{1}{2}}^{n}\,u_{j}^{n} - u_{j-1}^{n}}{\Delta x^2}$$

Eq. 2.20

$u_{tx} \rightarrow$ first-order upwind differencing

$(\nu(u)u_x)_x \rightarrow$ nonlinear centered Laplacian evaluated from u^n

Linear case: $\nu(u) = \nu_0$

$$\frac{u_{j}^{n+1} - u_{j-1}^{n+1} - u_{j}^{n} + u_{j-1}^{n}}{\Delta t_H \Delta x} = \nu_0 \frac{u_{j+1}^{n} - 2u_{j}^{n} + u_{j-1}^{n}}{\Delta x^2}$$

Rearranging gives the full nonlinear ordered SORh sweep

NONLINEAR SORh UPDATE

$$u_j^{n+1} = u_j^n + \omega_H[\underbrace{(u_{j-1}^{n+1} - u_{j-1}^n)}_{\text{already-computed upstream correction}} + \frac{\Delta t_H}{\Delta x}\{\nu^n_{j+\frac{1}{2}}(u^n_{j+1} - u^n_j) - \nu^n_{j-\frac{1}{2}}(u^n_j - u^n_{j-1})\}]$$

CONSTANT-COEFFICIENT REDUCTION

$$u_j^{n+1} = u_j^n + \omega_H[(u_{j-1}^{n+1} - u_{j-1}^n) + r_H \cdot S_j^n]$$

$$r_H = \frac{\nu_0 \Delta t_H}{\Delta x}, \qquad S_j^n = u_{j+1}^n - 2u_j^n + u_{j-1}^n, \qquad \Delta t_H = \mathrm{CFL}\frac{\Delta x}{\nu_{max}}$$

The correction form isolates how upstream information enters each sweep

Point Correction $\delta_j^{n+1} = u_j^{n+1} - u_j^n$

SORp

$$\delta_j^{n+1} = \omega_P \cdot \Delta t_P \cdot R_j^{GS}$$

residual injection + GS upstream corrections

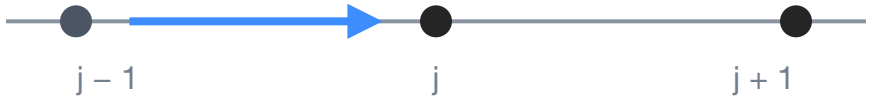
SORh

$$\delta_j^{n+1} = \omega_H[\delta_{j-1}^{n+1} + \Delta t_H \cdot \Delta x \cdot R_j^n]$$

explicit upstream propagation

residual injection

Sweep Dependency



→ In SORp, the updated upstream value u_{j-1}^{n+1} is embedded into the residual R_j^{GS}.

→ In SORh, the residual stays on u^n; the upstream value u_{j-1}^{n+1} is added separately through δ_{j-1}^{n+1}.

An O(Δx) hyperbolic timestep does not produce an O(Δx) elliptic correction

PARABOLIC

$$\Delta t_P = O(\Delta x^2)$$

$$\Delta t_P \cdot R = O(\Delta x^2)R$$

HYPERBOLIC

$$\Delta t_H = O(\Delta x)$$

$$\Delta t_H \cdot \Delta x \cdot R = O(\Delta x^2)R$$

↑ **extra Δx appears from the mixed derivative**

Both methods inject an elliptic correction at the same O(Δx²) scale.

Method differences appear in spectral radius, cost per sweep, parameter scaling, and other factors.

In correction form, AOR separates propagation from residual injection

Define: $A = D - L - U, \quad \delta^{n+1} = u^{n+1} - u^n$

Standard AOR iteration, rewritten in terms of the correction*:

$$(D - r_A L)\delta^{n+1} = \omega_A(b - Au^n)$$

upstream propagation

The propagation parameter controls the lower-triangular, Gauss–Seidel-like sweep.

residual injection

The residual parameter scales the newly introduced correction $b - Au^n$.

Classical SOR is the one-parameter line: $r_A = \omega_A$

*Appendix A.1 in the report shows how the AOR correction form is derived.

Linear SORh is AOR under a simple parameter map

CORRECTION FORMS

AOR

$$\delta_j^{n+1} = \frac{r_A}{2}\delta_{j-1}^{n+1} + \frac{\omega_A}{2}S_j^n$$

SORh

$$\delta_j^{n+1} = \omega_H \cdot \delta_{j-1}^{n+1} + \omega_H \cdot r_H \cdot S_j^n$$

MATCH THE COEFFICIENTS TERM BY TERM

UPSTREAM COEFFICIENT

$$\frac{r_A}{2} = \omega_H$$

$$\Rightarrow$$

$$r_A = 2\omega_H$$

ω_h sets upstream propagation strength.

RESIDUAL COEFFICIENT

$$\frac{\omega_A}{2} = \omega_H \cdot r_H$$

$$\Rightarrow$$

$$\omega_A = 2r_H \cdot \omega_H$$

$\omega_h \cdot r_h$ controls residual injection.

*The AOR/SOR identity applies only for $\nu(u) = \nu_0$, not to the state-dependent nonlinear iteration.

At unit CFL, SORh reduces to classical SOR

$$\mathrm{CFL}_H = r_H = 1 \quad \longrightarrow \quad r_A = \omega_A = \alpha = 2\omega_H$$

SHARED CLASSICAL-SOR CORRECTION

$$\delta_j^{n+1} = \frac{\alpha}{2}\delta_{j-1}^{n+1} + \frac{\alpha}{2}S_j^n$$

1D OPTIMAL RELAXATION PARAMETERS

CLASSICAL SOR

$$\alpha_{opt} = \frac{2}{1 + \sin(\pi/(M+1))}$$

SORh (M = N – 1)

$$\omega_{H,opt} = \frac{1}{1 + \sin(\pi/N)}$$

SORp

$$\omega_{P,opt} = \frac{2}{\mathrm{CFL}_P[1 + \sin(\pi/N)]}$$

Exact for constant coefficients at unit hyperbolic CFL; below unit CFL, SORh remains AOR rather than classical SOR.

The spectra confirm equivalence in the linear limit

LINEAR SPECTRUM

- CFL = 0.8
- $N_{int} = 64$

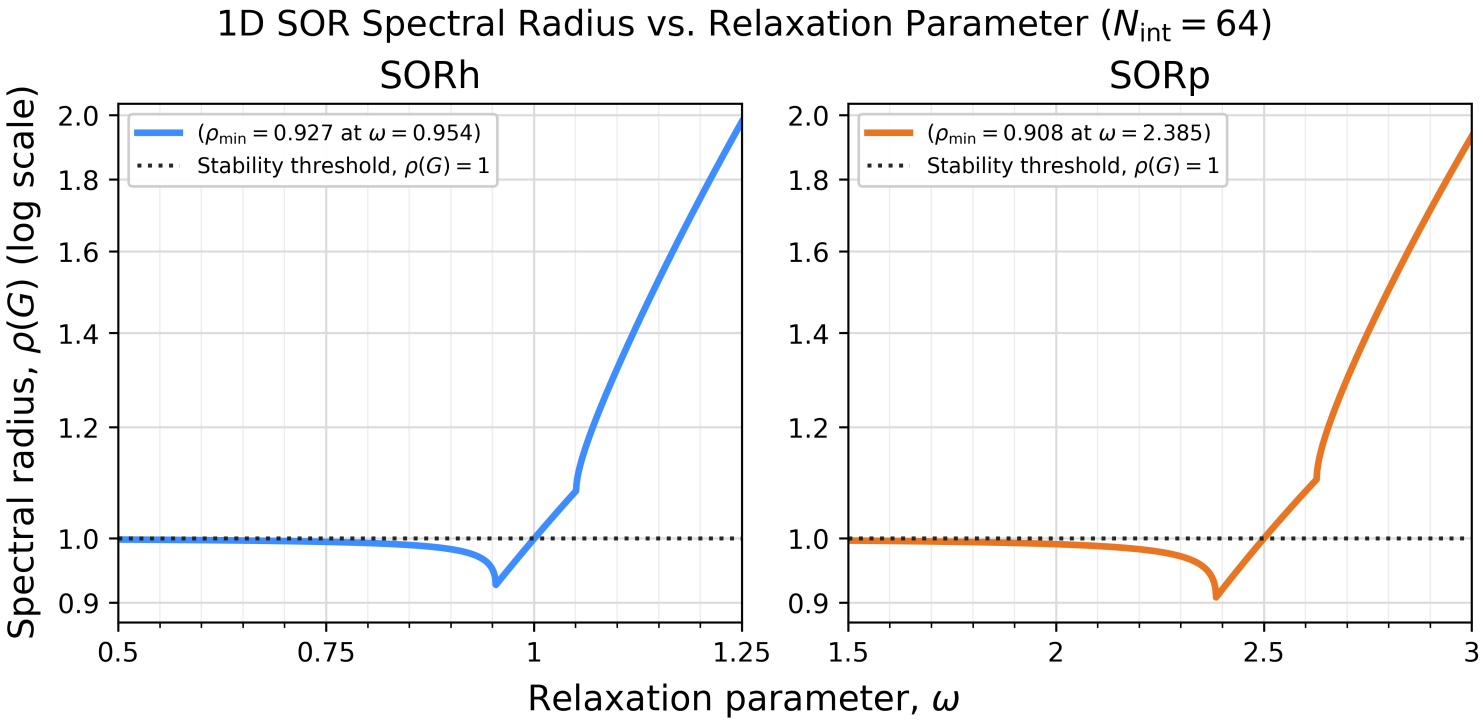
SORh

$$\rho_{min} \approx 0.9263$$

SORp

$$\rho_{min} \approx 0.9078$$

Measured optima agree with predicted ω_{opt} formulas.



On the linear problem, SORp retains a modest advantage

CONTROL CASE

- CFL = 0.8
- $\nu = 0.5$
- common tolerance

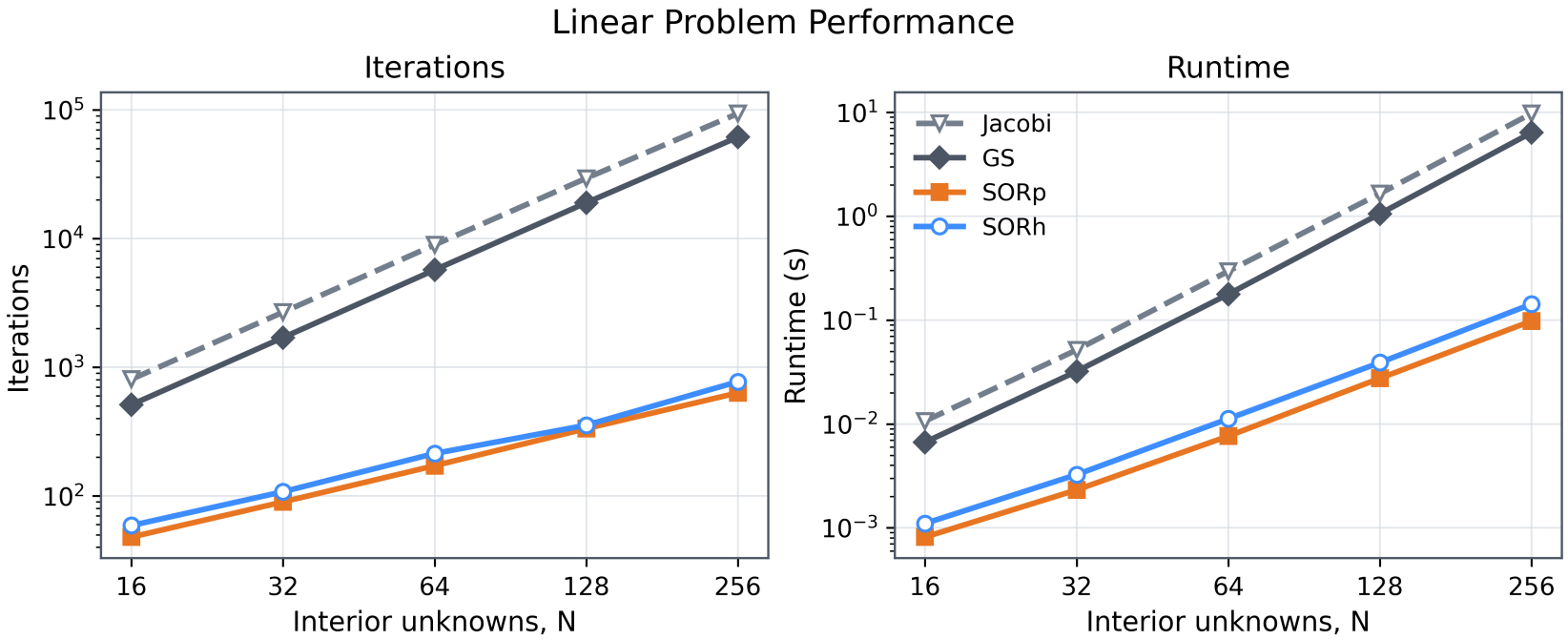
19–22%

more SORh iterations

~20% higher point-update cost

≈42%

longer runtime



Separating propagation and residual injection does not help the constant-coefficient problem.

On the nonlinear problem, SORh changes the scaling trend

GRID REFINEMENT

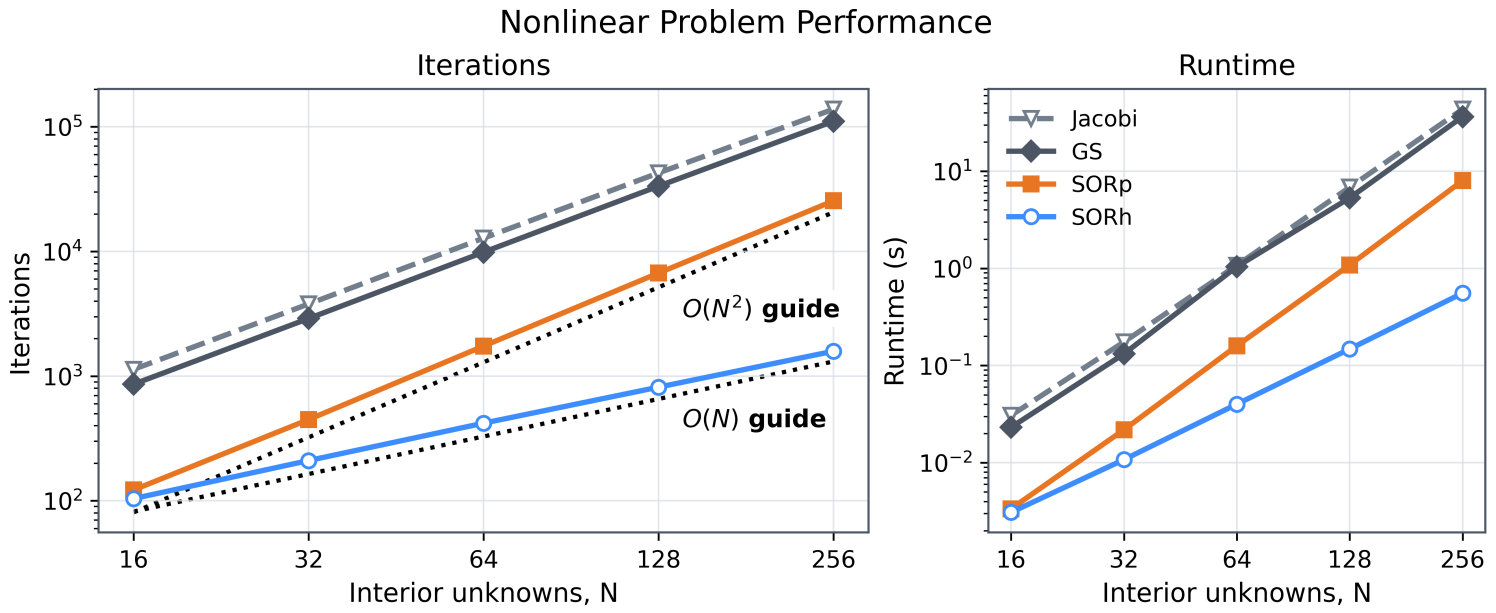
SORh

$$\approx O(N)$$

SORp

$$\approx O(N^2)$$

- **Parameters optimized independently for the nonlinear problem.**
- **Exact-solution diagnostics differ by only 0.57%.**



Iteration count is the primary comparison; runtime confirms the same separation.

Weak diffusion throttles SORp correction transport, but not SORh

SORp

$$\delta_j^{n+1} \approx \omega_P \frac{\Delta t_P \nu_{j-\frac{1}{2}}}{\Delta x^2} \delta_{j-1}^{n+1} + \Delta t_P R_j^n$$

upstream propagation × local ν

$\nu_{j-1/2} \to 0$



Local face diffusion narrows correction transport.

SORh

$$\delta_j^{n+1} = \omega_H \delta_{j-1}^{n+1} + \Delta t_H \Delta x R_j^n$$

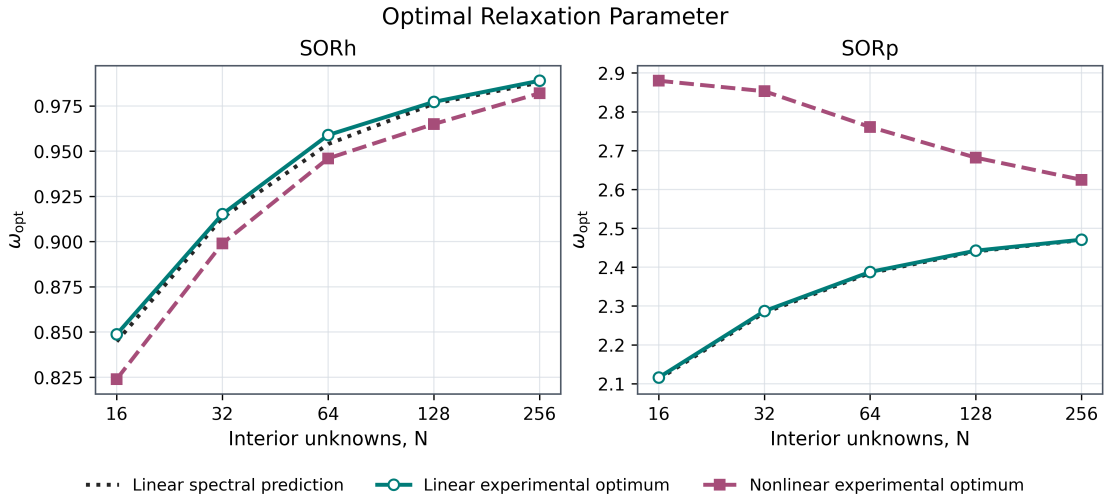
coefficient-independent upstream term

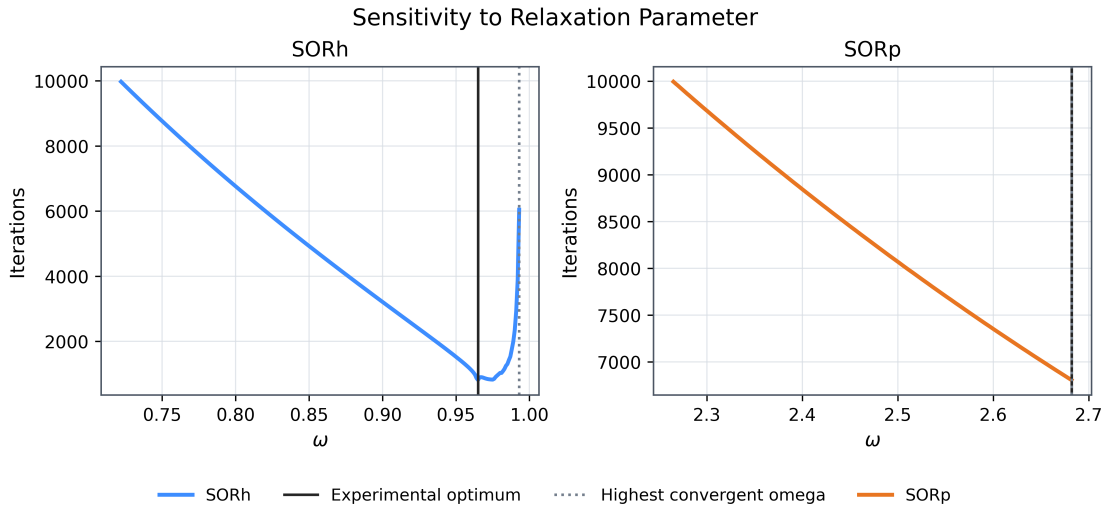


Explicit propagation remains intact through the weak edge.

A single global ω_p cannot strengthen the weak region without over-amplifying stronger-diffusion regions.

SORh remains predictably tunable without sacrificing steady-state accuracy





SORh: Optimum stays near the linear prediction.

SORp: Moves toward the stability boundary.

Roadmap: Transition to 2D Methods

1D FOUNDATION

hyperbolic derivation · AOR / SOR identity · weak-diffusion mechanism

MAIN 2D DEVELOPMENT

1

DIRECTIONAL MODEL

two transport directions

2

DISCRETE METHOD

full update and ordering

3

NUMERICAL RESULTS

linear Poisson + nonlinear diffusion

4

SELF-GRAVITY

Poisson solve in FLASH

block / wavefront execution

The 2D section develops the method, characterizes its behavior, and applies it to FLASH.

The two-dimensional tests connect SORh to self-gravity and variable diffusion

LINEAR POISSON TEST

$\rho(x, y)$

mass density

$\Delta\varphi = 4\pi G\rho$

gravitational potential

$g = -\nabla\varphi$

acceleration field used by the application

NONLINEAR DIFFUSION TEST

$$\nabla \cdot (\nu(u)\,\nabla u) = 0$$

$$\nu(u) = \frac{u}{1+u}$$

$$\nu(u) \to 0 \quad \text{as} \quad u \to 0$$

Poisson tests the linear classical limit; variable diffusion tests the weak-local-coupling mechanism.

Single-Direction 2D Transport Model

1 REFERENCE 1D HYPERBOLIC MODEL

$$\varphi_{tx} = \varphi_{xx} - f\,.$$

2 DEFINE A TRANSPORT DIRECTION

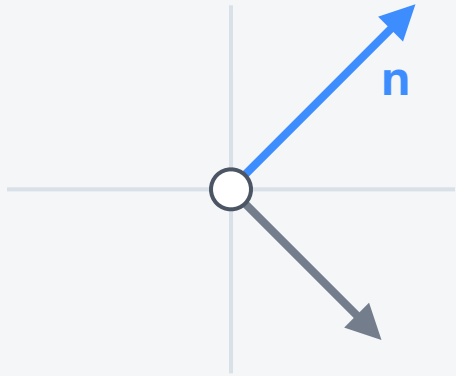
$$n = (n_1, n_2) \quad \rightarrow \quad \partial_n := n \cdot \nabla = n_1\partial_x + n_2\partial_y$$

3 APPLY IT TWICE

$$(\partial_n)^2 = n_1^2\partial_{xx} + 2n_1n_2\partial_{xy} + n_2^2\partial_{yy}$$

4 2D SINGLE-DIRECTION HYPERBOLIC MODEL

$$\partial_t\partial_n\varphi = (\partial_n)^2\varphi - f$$



curvature retained
along one axis

missing orthogonal curvature

$$(\partial_n)^2 \neq \Delta$$

Rank-one diffusion operator cannot reproduce 2D Laplacian.

Two orthogonal directions recover the Laplacian and select one transport axis

1 RESTORE THE MISSING CURVATURE

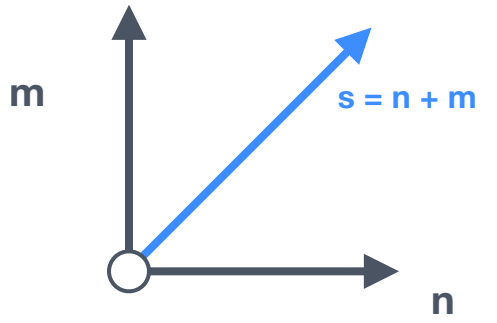
$$\partial_t(\partial_n + \partial_m)\varphi = [(\partial_n)^2 + (\partial_m)^2]\varphi - f$$

2 ORTHONORMAL RESTRICTION

$$n \perp m \Rightarrow (\partial_n)^2 + (\partial_m)^2 = \Delta$$

3 CHOOSE A DIRECTION

$$s = n + m, \partial_s = \partial_n + \partial_m$$



$$n = (1, 0), m = (0, 1), s = (1, 1)$$

PARABOLIC — SORp

$$\varphi_t = \Delta\varphi - f$$

HYPERBOLIC — SORh

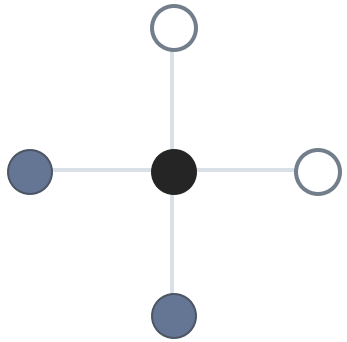
$$\varphi_{tx} + \varphi_{ty} = \Delta\varphi - f$$

Just like 1D: Same elliptic steady target; different pseudo-time dynamics.

The 2D discretization combines upwind transport with a five-point residual

$\varphi_{tx} + \varphi_{ty} \rightarrow$ **BACKWARD DIFFERENCES**

$$\frac{(D_x^- \varphi^{k+1})_{i,j} - (D_x^- \varphi^k)_{i,j}}{\Delta t_H} + \frac{(D_y^- \varphi^{k+1})_{i,j} - (D_y^- \varphi^k)_{i,j}}{\Delta t_H} = R_{i,j}^k$$



west / south are already updated

FIVE-POINT ELLIPTIC RESIDUAL

$$R_{i,j}^k = \frac{\varphi_{i+1,j}^k - 2\varphi_{i,j}^k + \varphi_{i-1,j}^k}{h_x^2} + \frac{\varphi_{i,j+1}^k - 2\varphi_{i,j}^k + \varphi_{i,j-1}^k}{h_y^2} - f_{i,j}$$

PSEUDO-TIME CONVENTIONS

SORp $\Delta t_P = \dfrac{CFL_P}{2(h_x^{-2} + h_y^{-2})} = O(h^2)$

SORh $\Delta t_H = CFL_H \min(h_x, h_y)$

SORh exposes the upstream corrections, SORp embeds them in its residual

SORp — PARABOLIC REFERENCE

$$\varphi_{i,j}^{k+1} = \varphi_{i,j}^{k} + \omega_P \cdot \Delta t_P \cdot R_{i,j}^{GS,k}$$

Updated west and south neighbors enter through the Gauss–Seidel residual.

SORh — FULL UNSPLIT UPDATE

$$\varphi_{i,j}^{k+1} = \varphi_{i,j}^{k} + \omega_H c_1 [h_y(\varphi_{i-1,j}^{k+1} - \varphi_{i-1,j}^{k}) + h_x(\varphi_{i,j-1}^{k+1} - \varphi_{i,j-1}^{k}) + c_2 R_{i,j}^{k}]$$

$$c_1 = \frac{1}{h_x + h_y}, \qquad c_2 = \Delta t\, h_x\, h_y, \qquad h_x = \Delta x, h_y = \Delta y, \qquad \Delta t_H = \mathrm{CFL}_H \min(h_x, h_y)$$

In a forward lexicographic sweep, SORh writes the two upstream corrections explicitly; the residual remains on the old iterate.

The linear AOR/SOR map survives in two dimensions

AOR

$$\delta_{i,j}^{k+1} = \frac{r_A}{4}(\delta_{i-1,j}^{k+1} + \delta_{i,j-1}^{k+1}) + \frac{\omega_A}{4} h^2 R_{i,j}^{k}$$

SORh

$$\delta_{i,j}^{k+1} = \frac{\omega_H}{2}[\delta_{i-1,j}^{k+1} + \delta_{i,j-1}^{k+1} + CFL_H \cdot h^2 R_{i,j}^{k}]$$

ROLE	AOR COEFFICIENT	SORh COEFFICIENT	MAPPING
UPSTREAM PROPAGATION	$\frac{r_A}{4}$	$\frac{\omega_H}{2}$	$r_A = 2\omega_H$
RESIDUAL INJECTION	$\frac{\omega_A}{4}$	$\frac{\omega_H \cdot CFL_H}{2}$	$\omega_A = 2\omega_H \cdot CFL_H$
CLASSICAL SOR	$r_A = \omega_A = \alpha$	$CFL_H = 1$	$\alpha = 2\omega_H$

The optimal relaxation parameter formulas are unchanged* from 1D, because:

- **The 2D transport normalization compensates for the larger diagonal factor.**
- **The Jacobi spectral radius and classical optimum are unchanged.**

*Formulas are exact at unit CFL on a uniform M × M square grid with the centered five-point stencil.

SORh SQUARE-GRID FORMULA

$$\omega_{H,opt}^{2D} = \frac{1}{1 + \sin(\pi/N)}, N = M + 1$$

Nonlinearity changes the residual, not the upstream transport structure

1 OLD-ITERATE FACE COEFFICIENT

$$\nu^{k}_{i+1/2,j} = \nu\left(\frac{u^{k}_{i+1,j} + u^{k}_{i,j}}{2}\right)$$

Other faces are constructed analogously.

2 NONLINEAR FLUX-DIVERGENCE RESIDUAL

$$R^{NL,k}_{i,j} = \frac{F^{x}_{i+1/2,j} - F^{x}_{i-1/2,j}}{h_x} + \frac{F^{y}_{i,j+1/2} - F^{y}_{i,j-1/2}}{h_y}, \qquad F^{x} = \frac{\nu \Delta_x u}{h_x}, \quad F^{y} = \frac{\nu \Delta_y u}{h_y}$$

3 ORDERED SORh UPDATE

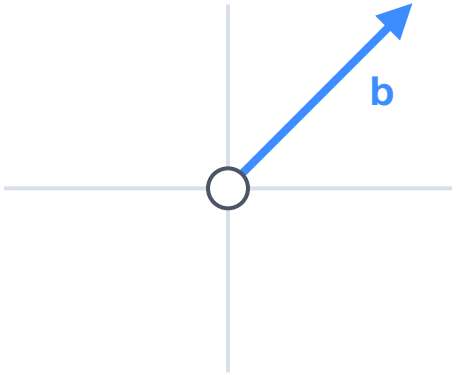
$$u^{k+1}_{i,j} = u^{k}_{i,j} + \omega_H c_1 [h_y \delta^{k+1}_{i-1,j} + h_x \delta^{k+1}_{i,j-1} + c_2 R^{NL,k}_{i,j}]$$

The face coefficients and residual change with u; the explicit west/south transport shell does not.

Directional SORh: Discretizing the Directional Transport Model

$$b = (b_x, b_y) \qquad p_x = \frac{|b_x|}{|b_x| + |b_y|} \qquad b_x\varphi_{tx} + b_y\varphi_{ty} = \Delta\varphi - f \qquad p_y = \frac{|b_y|}{|b_x| + |b_y|} \qquad p_x + p_y = 1$$

GRID DIAGONAL

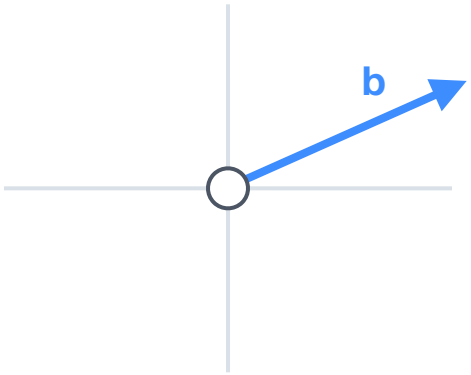


$$p_x = p_y = \frac{1}{2}$$

default SORh

west and south are upstream

OFF-DIAGONAL

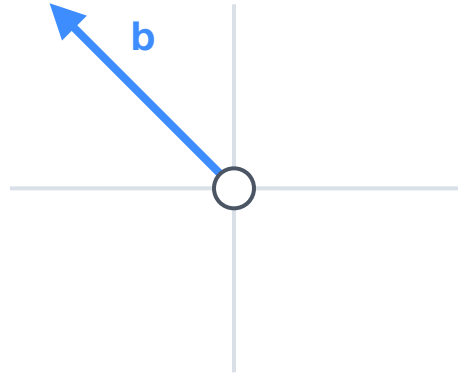


$$p_x > p_y$$

weighted x/y propagation

magnitudes tune the two pathways

MIRRORED QUADRANT



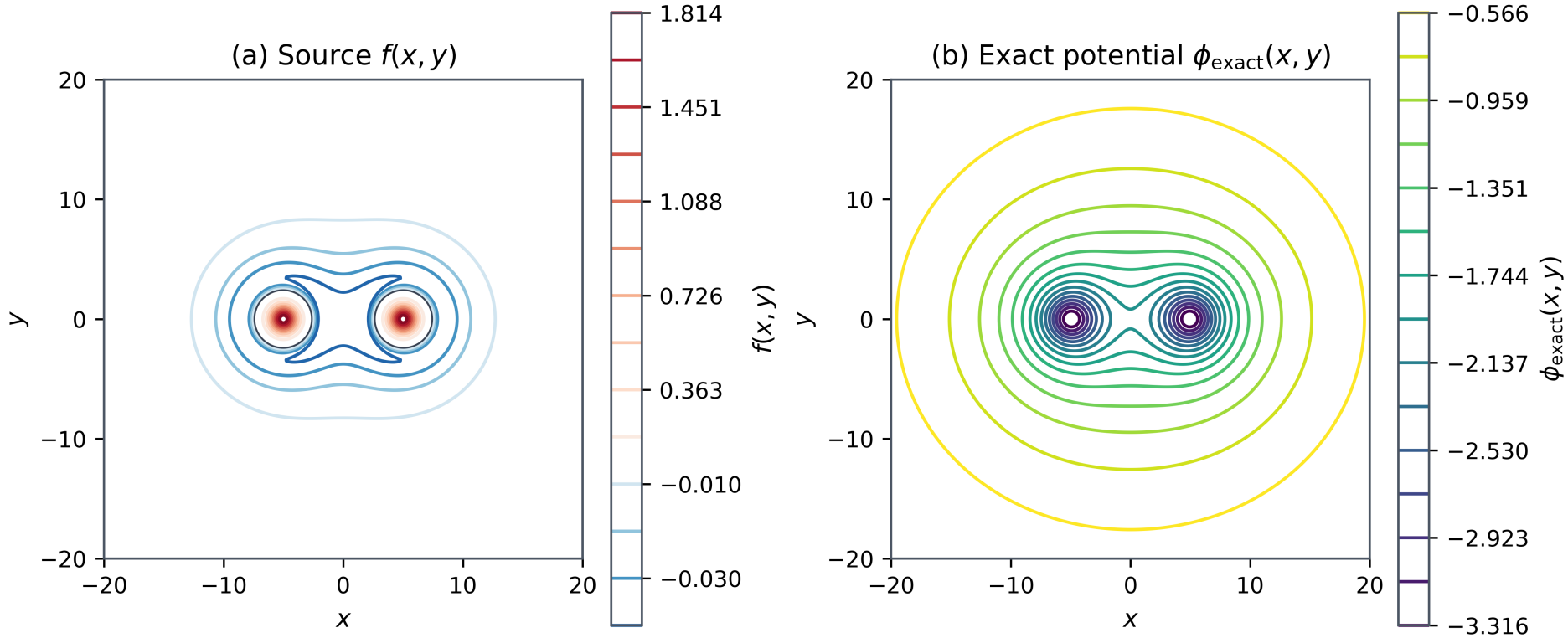
$$p_x = p_y = \frac{1}{2}$$

mirrored ordering

signs select east and south upstream

A genuine directional generalization, but showed no significant practical advantage in the tested cases.

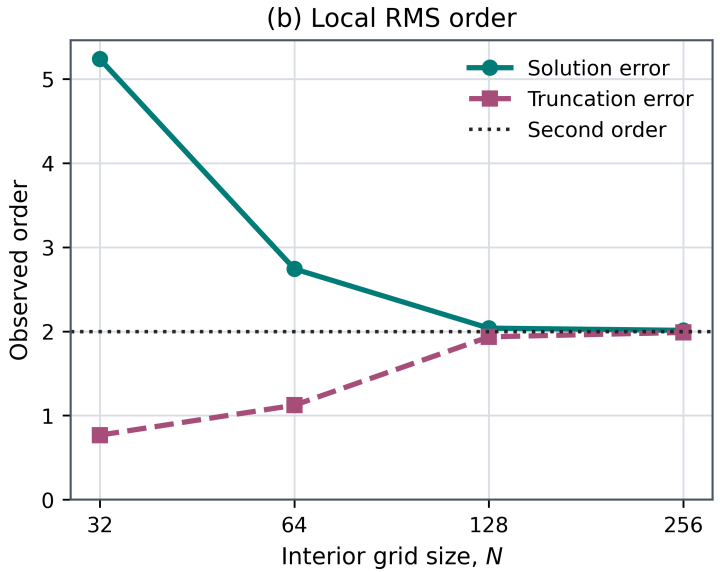
The analytic ‘two-blob’ Poisson problem



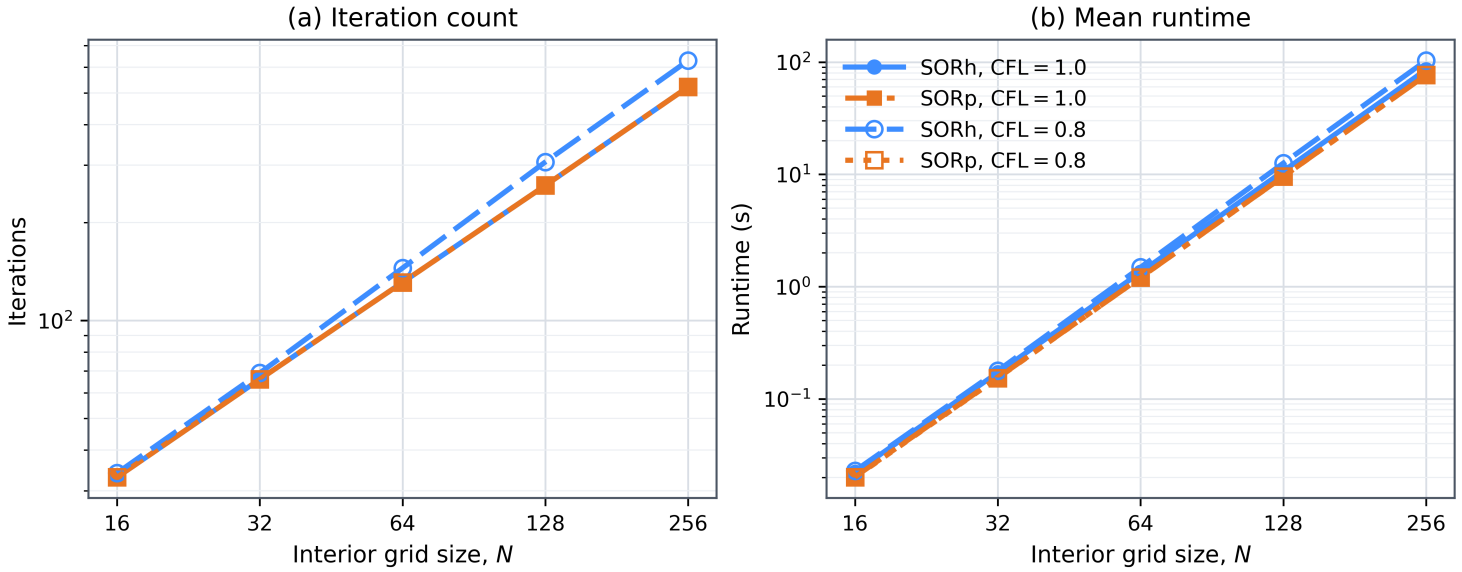
$$\phi_{\text{exact}}(x, y) = -4\pi G\rho_0 R^2 \left[1 + \frac{(x-5R)^2 + y^2}{3R^2}\right]^{-1/2} + \left[1 + \frac{(x+5R)^2 + y^2}{3R^2}\right]^{-1/2}, \quad R = 1, \quad \rho_0 = 3, \quad 4\pi G = 1$$

The 2D linear problem reproduces the classical-SOR baseline

Accuracy approaches 2nd order



Nearly identical performance



33 → 522
sweeps for both methods

$k \propto N^{0.995}$
near-linear sweep scaling

- **SORp is slightly cheaper per sweep**
- **Iteration counts are identical for CFL = 1**
- **Both converge to same discrete solution**

The nonlinear 2D problem amplifies SORh's sweep advantage

GRID REFINEMENT

SORh

139 → 662

SORp

221 → 3502

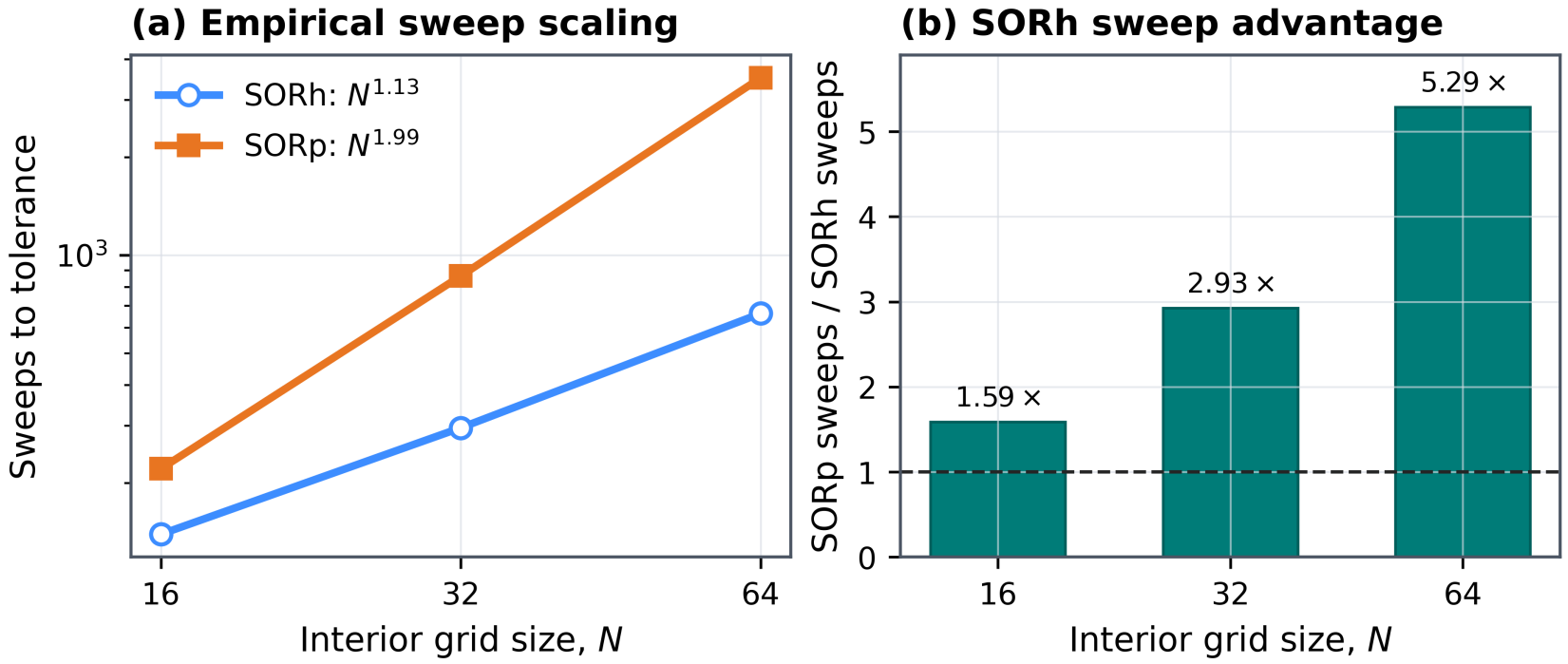
SWEEP ADVANTAGE

1.59× → 5.29×

DESCRIPTIVE FITS

SORh: $N^{1.13}$

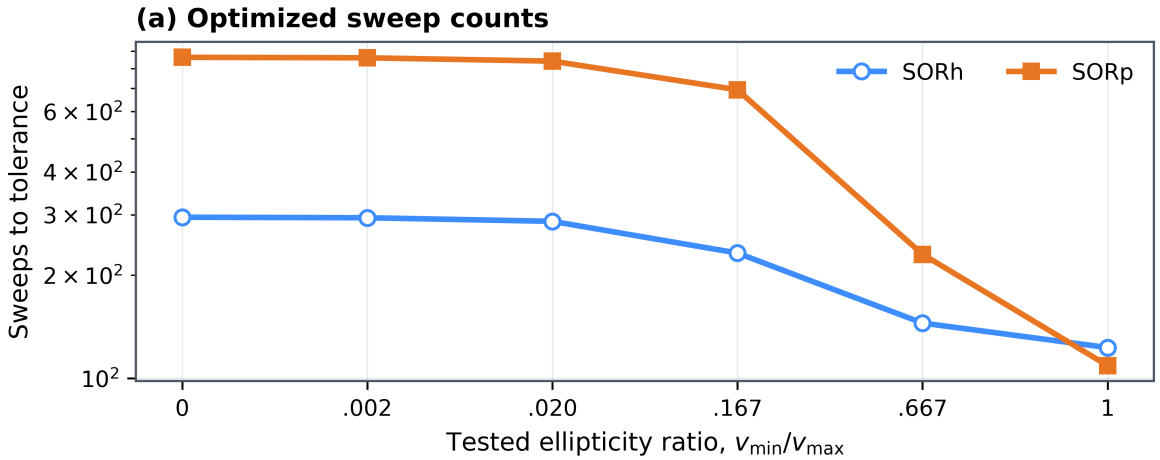
SORp: $N^{1.99}$

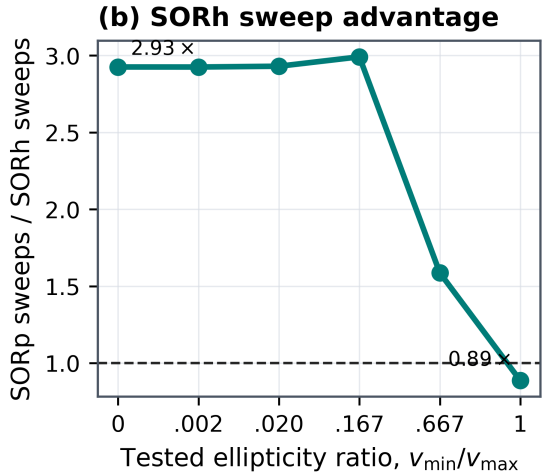


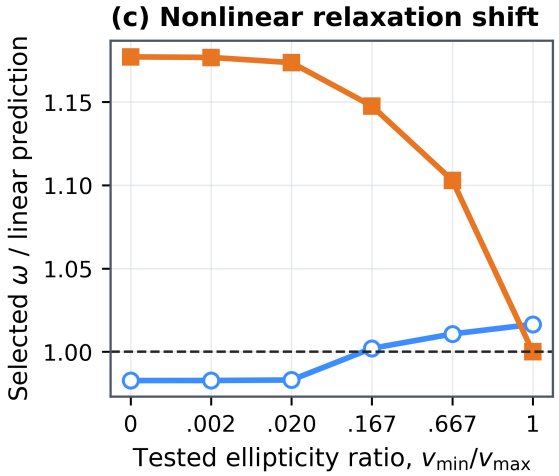
The sweep separation grows with refinement; runtime follows the same trend.

Fit over limited grid refinement range. Converged solutions agree to ≈10^{-6} in normalized error norm.

The advantage tracks coefficient contrast, not strict degeneracy alone







Direct mechanism evidence, not a nonlinear spectral proof.

UPSTREAM-CORRECTION WEIGHTS

$c_H = \omega_H/2, \quad c_{P,min} = 0.4\omega_P\nu_{min}$

minimum observed $c_H / c_{P,min}$ (unregularized)

N	$c_H / c_{P,min}$
16	**7.3**
32	**13.9**
64	**26.5**

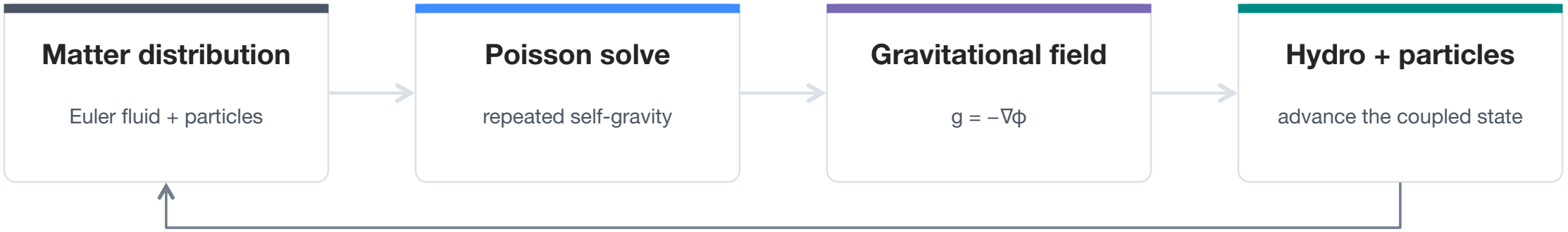
WHAT THE CONTROLS SHOW

- **mild regularization** — advantage persists
- **more uniform ν** — advantage declines
- **constant coefficient** — ordering reverses

Coefficient contrast is a strong predictor of the separation.

The FLASH Pancake test asks whether SORh survives production coupling

ZEL'DOVICH PANCAKE · PERIODIC COSMOLOGICAL DENSITY PERTURBATION



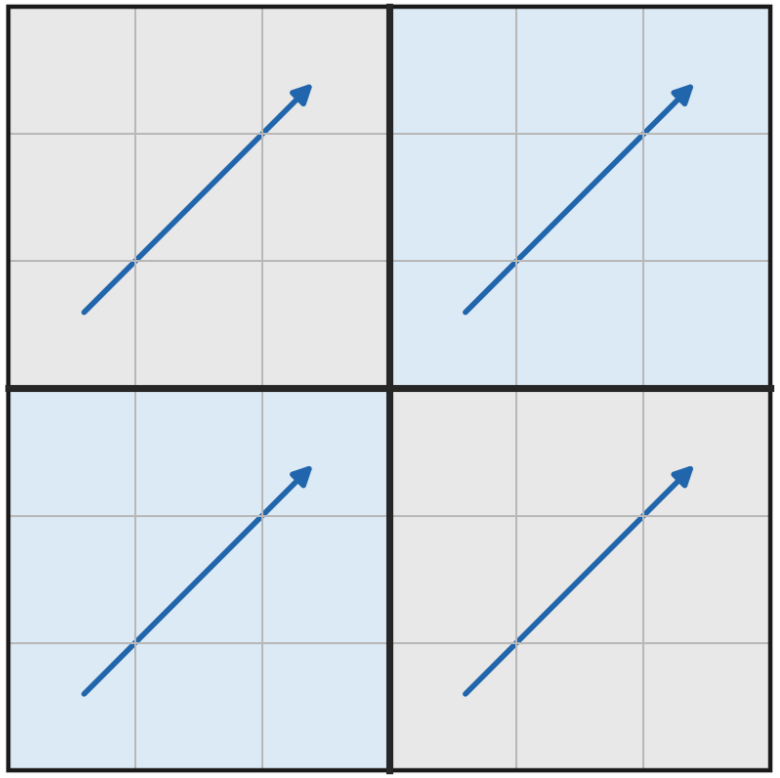
PRIMARY FIXED-MESH CASE

128 × 128	64 blocks	1 and 4	Huang–Greengard	10^{-6}	$m_{max} = 200$
uniform mesh	16 × 16 cells each	MPI ranks	Multigrid reference	shared solver tolerance	SORh iteration cap

The fixed mesh isolates solver performance and ordering behavior.

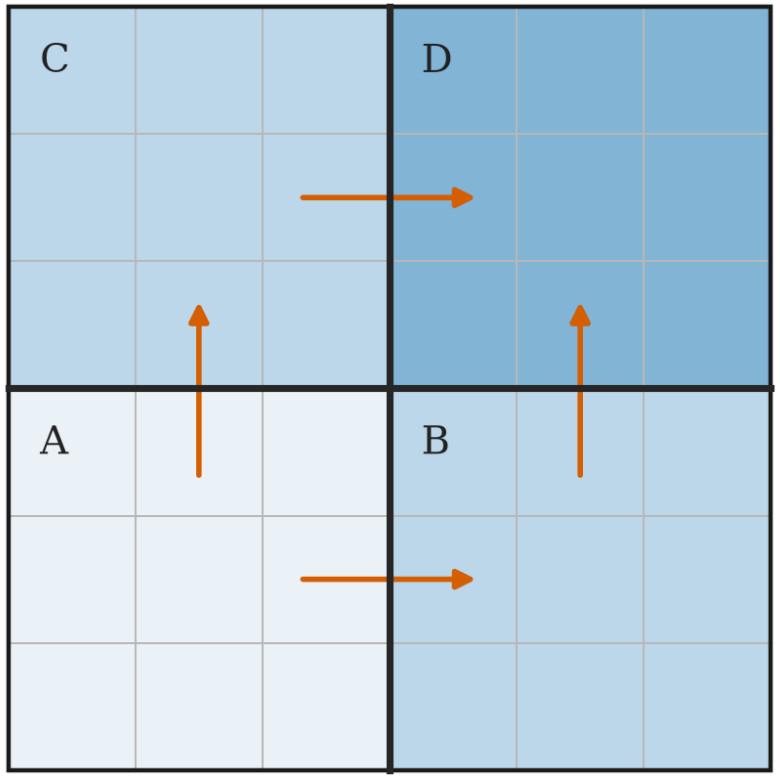
Block and wavefront modes represent different global iterations

(a) Block mode: independent local sweeps



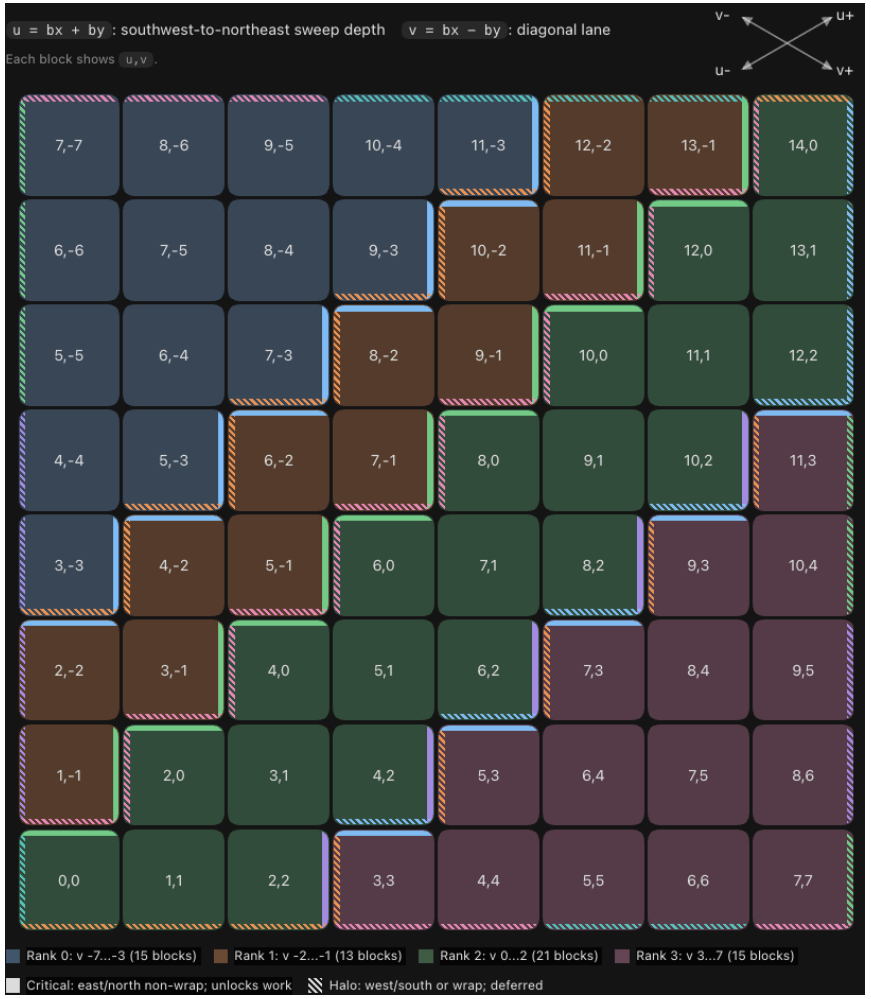
Concurrent blocks use lagged values across block boundaries

(b) Wavefront mode: west/south block dependencies



$A \to \{B, C\} \to D$; readiness is asynchronous

Wavefront mode: preserve the ordered iteration while exposing diagonal concurrency



$$u = b_x + b_y$$

sweep depth · equal-u blocks form one ready front

$$v = b_x - b_y$$

diagonal lane · contiguous v-ranges define rank ownership

WEST / SOUTH READY

A block starts only after both current-sweep upstream faces arrive.

CROSS-RANK FACE PUBLICATION

Completed faces are published immediately so the next diagonal can advance.

Rank layout changes execution and communication, but the original sweep ordering is preserved.

With 200 iterations, wavefront SORh is competitive with MG on four ranks

CAP BUDGET

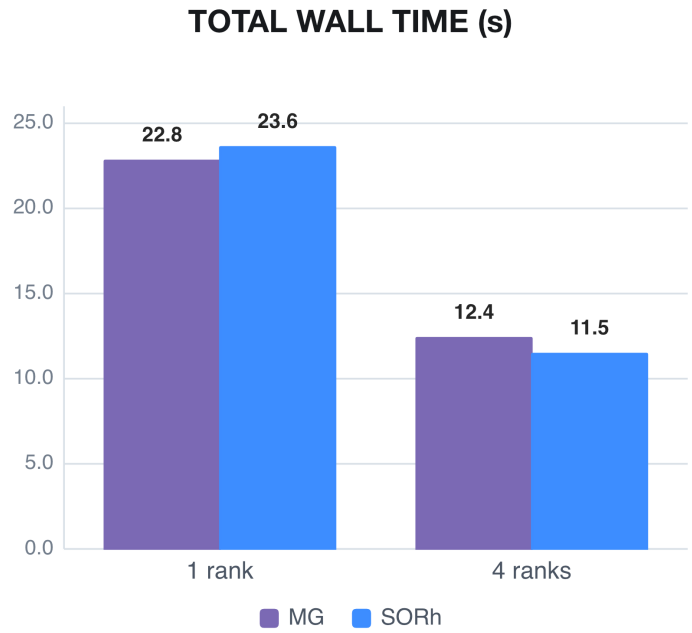
times relative to MG · final profile difference

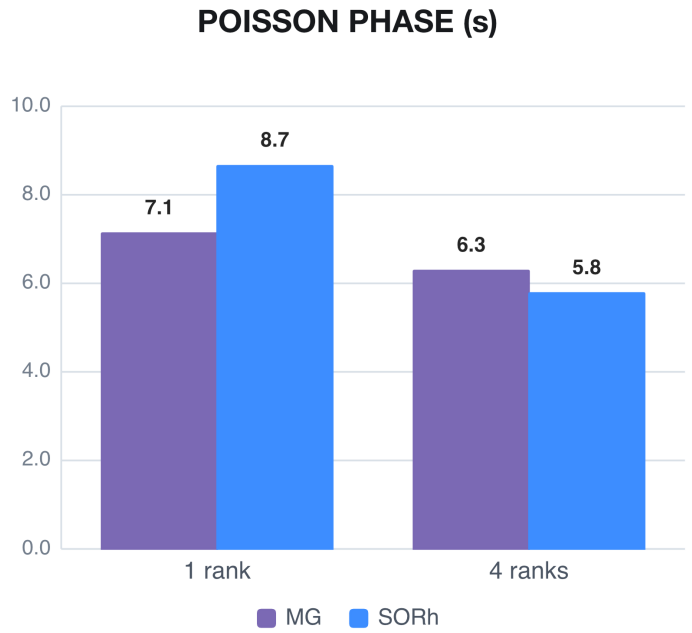
m_{max}	Sweeps Converged	Wall	Poisson	final Δ
100	21/419	0.729	0.494	17.207%
200	**34/417**	**0.925**	**0.919**	**0.228%**
300	48/417	1.133	1.337	0.224%
400	214/417	1.345	1.754	0.004%
500	372/417	1.415	1.895	0.002%

$m_{max} = 200$ · **BEST COMPROMISE**

34 / 417	0.925	0.228%
converged steps	SORh / MG Wall-clock time	max final five-field Δ

PRIMARY COMPARISON: $m_{max} = 200$

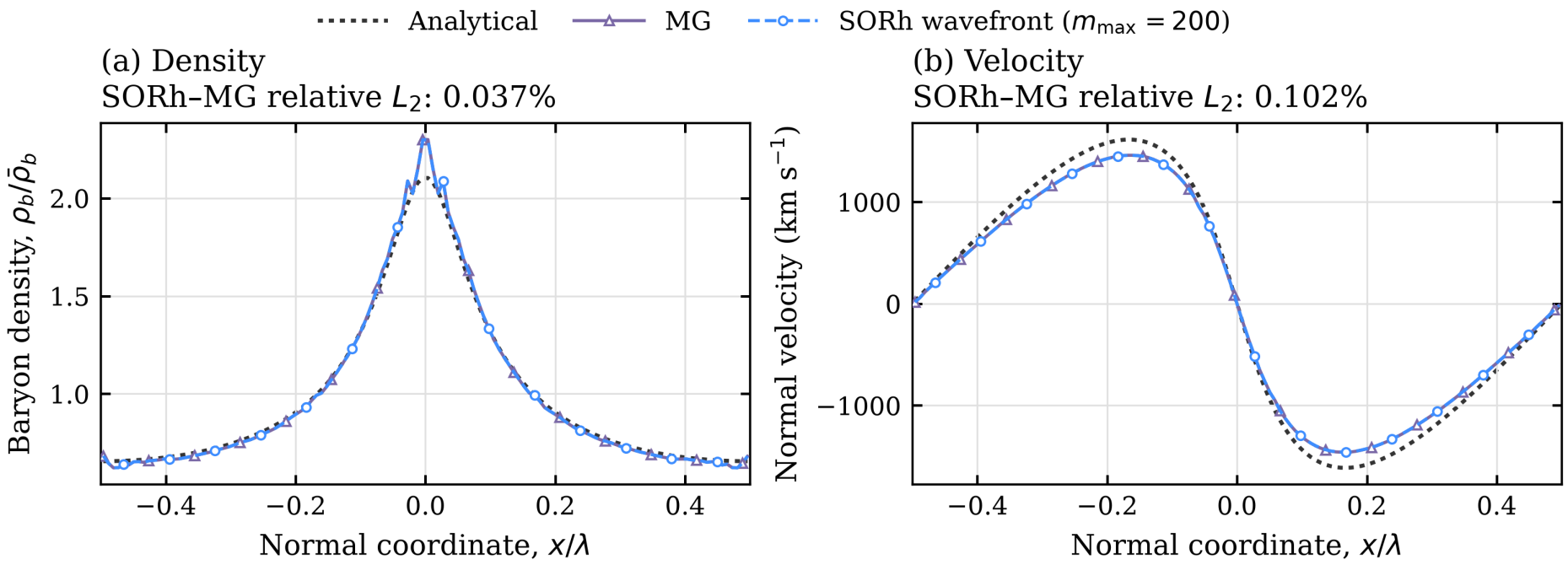




4 ranks, $m_{max} = 200$: **7.5%** faster wall **8.1%** faster Poisson

On one rank, SORh is slightly slower; the ordering benefit appears in the four-rank execution.

Cost–accuracy tradeoff: SORh iteration cap



Direct SORh–MG difference: 0.0367% density · 0.1018% velocity

100	$m_{max} = 200$	400–500
fast, inaccurate	**best compromise**	slow, near-MG accuracy

The shared hydro/discretization error is much larger than the solver-to-solver profile difference.

Summary of Results

REGIME	WHAT THE RESULTS SAY	EVIDENCE BOUND
LINEAR · CONSTANT COEFFICIENT	**SORh is AOR/SOR; there is no inherent sweep advantage.**	SORh-AOR parameter map Same classical limit
WEAK, STRONGLY NONUNIFORM DIFFUSION	**Explicit correction transport reduces sweeps when local coupling weakens.**	1D: $O(N)$ vs. $O(N^2)$ scaling 2D: up to 5.29× fewer sweeps
FLASH · FIXED-MESH PANCAKE	**Wavefront SORh is viable with a controlled inexact solve.**	4 ranks: 7.5% faster wall time at $m_{max} = 200$

SCOPE No general nonlinear theory · limited directional evidence · uniform FLASH mesh · 1–4 ranks

Topics for Future Work

MOTIVATING QUESTION

How should SORh generalize beyond the tested setting?

STENCIL COMPATIBILITY + ALGEBRAIC DESIGN

Determine when mixed-derivative and elliptic stencils produce a prescribed AOR/SOR splitting.

NONLINEAR THEORY + METHOD COMPARISONS

Develop coefficient-contrast theory; optimize two-parameter AOR; compare Nishikawa-style formulations on nonuniform and anisotropic grids.

PRODUCTION BREADTH + PARALLEL SCALE

Test smoother and multigrid roles, true AMR, 3D, additional FLASH problems, and wider parallel scaling.

Three conclusions of SORh

1 — **Hyperbolic transport gives AOR's two parameters a physical interpretation.**

2 — **Separating correction propagation from residual injection matters when weak local diffusion throttles a sweep.**

3 — **Wavefront ordering preserves the global iteration and can be competitive in a scoped production test.**

Questions